Axiom-Preserving Differential Extensions and Proper Two-Forms

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Abstract. This note formalizes a framework for extending a fixed differential function from a base field to a field extension while preserving a declared package of structural axioms. The resulting extended function is then used to produce an overcomplete two-parameter representation $x = r\tilde{g}(\iota k)$. The central result gives necessary and sufficient coverage, non-uniqueness, and branch-selection conditions for this representation to be a proper two-form in the sense of the accompanying n-form framework. A real-to-complex exponential example recovers polar form as a canonical model.

1. Purpose and scope

The goal is to separate four layers that are often conflated: a base field, a field extension, a field of functions over the base, and a field of functions over the extension. The framework then asks when a fixed differential function can be extended by preserving its defining axioms and when that extension yields a proper two-form.

The operation $\oplus$ is retained as part of the generalized extension data. It may control closure, generation, or other structural rules of the extension. The principal formula below uses the ordinary multiplication of the ambient field; consequently, $\oplus$ affects the theorem only through the axiom package used to construct the extension, unless $\oplus$ is inserted explicitly into the formula.

2. Typed ambient data

Let F be a field of characteristic zero and let E be a field extension of F. Let ι be a designated nonzero element of E. Equip E with an additional operation $\oplus$, possibly subject to a separate closure or generation theory:

$$F \subseteq E, \quad \iota \in E^*, \quad \oplus : E \times E \to E$$

Let K be a differential field whose elements are functions defined over F, with derivation $D : K \to K$. Let J be a differential field of functions defined over E, with derivation $\tilde{D} : J \to J$. An extension map $\mathrm{Ext} : K \to J$ is called differential when it is a field embedding and intertwines the derivations:

$$\tilde{D} \circ \mathrm{Ext} = \mathrm{Ext} \circ D$$

A fixed element g in K is regarded as a function. Its extended function is $\tilde{g} = \mathrm{Ext}(g)$ in J. Function evaluation is part of the structure: whenever k belongs to the allowed parameter set $A \subseteq F$, the value $\tilde{g}(\iota k)$ is required to be defined in E.

3. Axiom-preserving extension of the derivative

The usual derivation laws are the minimum part of the extension package:

- $\tilde{D}(a + b) = \tilde{D}(a) + \tilde{D}(b)$
- $\tilde{D}(ab) = \tilde{D}(a)b + a\tilde{D}(b)$
- $\tilde{D}|_K = D$

They are not, by themselves, intended to exhaust the behavior of a particular derivative. Let A_D be a declared collection of further axioms: for example, a chain rule for a chosen composition operation, a specified constant field, commutation with an automorphism group, or a normalization such as $D(t) = 1$. The extension is admissible only if the corresponding lifted axioms hold in J.

Definition 3.1 (admissible differential extension). An admissible extension of (K, D) is a differential embedding $\mathrm{Ext} : (K, D) \to (J, \tilde{D})$ for which every axiom in the declared extension package A_D^{ext} holds after replacing K and D by J and $\tilde{D}$. The package must be consistent with the chosen extension J.

In particular, the differential-eigenfunction condition is preserved automatically.

Proposition 3.2. If $D(g) = g$ and Ext is a differential embedding, then $\tilde{D}(\tilde{g}) = \tilde{g}$.

$$\tilde{D}(\tilde{g}) = \tilde{D}(\mathrm{Ext}(g)) = \mathrm{Ext}(D(g)) = \mathrm{Ext}(g) = \tilde{g}$$

This proposition does not assert that the image values of $\tilde{g}$ cover the extension. Coverage is a separate condition on the pair $(\tilde{g}, A)$.

4. The proper two-form

Because g is fixed rather than variable, the expression has two input coordinates. Write $F^* = F \setminus \{0\}$ and $E^* = E \setminus \{0\}$. Let A be a chosen subset of F, and define:

$$h : A \to E^*, \quad h(k) = \tilde{g}(\iota k)$$

Assume $h(k)$ is nonzero for every $k \in A$. Define the parameterization:

$$\Phi : F^* \times A \to E^*, \quad \Phi(r, k) = rh(k) = r\tilde{g}(\iota k)$$

This definition uses ordinary multiplication in E and the inclusion $F \subseteq E$. It is therefore meaningful even when ι is not the standard imaginary unit.

Theorem 4.1 (coverage criterion). The map Φ is surjective onto E^* if and only if $E^* = F^*h(A)$. Equivalently, $h(A)$ meets every multiplicative coset of F^* in E^*.

Proof. If $x = \Phi(r, k)$, then $x \in F^* h(A)$. Conversely, if $x \in F^* h(A)$, write $x = rh(k)$ with $r \in F^*$ and $k \in A$. Then $x = \Phi(r, k)$. ■

Theorem 4.2 (properness criterion with choice branches). Assume the coverage criterion. The map Φ is a proper two-form in the choice-function sense when, for every coset C of F^* in E^*, the intersection $h(A) \cap C$ contains at least two distinct values, and a section $\beta : h(A) \to A$ has been chosen with $h(\beta(u)) = u$ for every $u \in h(A)$.

Proof sketch. For fixed $x \in E^*$, the allowed k-values are precisely those for which $h(k)$ lies in the coset xF^*. Two distinct values u and v of h in that coset give two distinct parameters k and l. Their corresponding r-values are x/u and x/v, which are distinct because u and v are distinct. Thus neither coordinate is uniquely determined by x. The choice section gives a single canonical k-value whenever x/r lies in $h(A)$.

The coordinate-isolation rules are defined on compatible input data:

$$G_r(x, k) = \frac{x}{h(k)}$$

$$G_k(x, r) = \beta\left(\frac{x}{r}\right)$$

The first rule returns the exact r-coordinate of a compatible tuple. The second returns the canonical k-coordinate selected by β; it need not equal every originally supplied k-coordinate. This is precisely the principal-branch interpretation of omni-directional invertibility.

5. A canonical model: polar form

The standard exponential extension supplies a concrete nontrivial model. Let $F = \mathbb{R}$ and $E = \mathbb{C}$. Let K be a real differential function field containing $\exp(t)$, let J be its complex counterpart containing $\exp(z)$, and let D and $\tilde{D}$ be differentiation with respect to t and z, respectively. The extension map sends $\exp(t)$ to $\exp(z)$:

$$g(t) = \exp(t), \quad D(g) = g, \quad \tilde{g}(z) = \exp(z), \quad \tilde{D}(\tilde{g}) = \tilde{g}$$

Choose ι to be the usual complex i and take $A = [0, 2\pi)$. Then:

$$h(k) = \exp(ik)$$

The function h is injective on A and its image is the unit circle. Every nonzero complex number has a polar representation:

$$z = r\exp(ik), \quad r \in \mathbb{R}^*, \quad k \in [0, 2\pi)$$

Thus $\mathbb{C}^* = \mathbb{R}^* h(A)$. Moreover, every $\mathbb{R}^*$-coset in $\mathbb{C}^*$ meets the unit circle in two values, u and $-u$. Those two values produce distinct r- and k-representations, while the interval $[0, 2\pi)$ supplies a canonical angle. This is a proper two-form in the stated choice-branch sense.

6. Relation to the generalized operation $\oplus$

The construction above is intentionally modular. We define the operation $\oplus$ where E is an extension field generated by appending a non-real unit ι to F. The operation is governed by the following structural rules to ensure proper dimensional mapping and absolute closure:

Axiom (Unique Projection). For any $x, y \in F$, there exists a unique action by the extension unit ι such that:

$$\iota \oplus x = y$$

Furthermore, for all $z \in F$, $z \oplus x \neq y$, guaranteeing that the operation by the non-real element creates a mapping strictly unobtainable within the base field F.

Axiom (Absolute Closure). The operation of any two elements from the base field strictly maps into the two-dimensional extension space spanned by 1 and ι. For all $x, y \in F$, there exist $a, b \in F$ such that:

$$x \oplus y = a + b\iota$$

To make $\oplus$ contribute directly to the theory of $\tilde{g}$, these rules governing $\oplus$ are formally appended to the extension's axiom package, A_D^{ext}. By incorporating the absolute closure rule into A_D^{ext}, the differential operator $\tilde{D}$ is constrained to respect the dimensional bounds of the extension space.

7. Limits and research questions

The coverage requirement $E^* = F^* h(A)$ is restrictive. It is not implied by $D(g) = g$ and should be treated as a separate hypothesis or as a theorem to establish for a selected class of extensions. It also yields the cardinal bound $|E^*/F^*| \leq |A| \leq |F|$.

A second issue is the design of A_D^{ext}. Requiring every axiom true in K to hold in J may be inconsistent. The appropriate object is a specified extension-stable package, together with an existence theorem or a counterexample for the proposed extension.

Finally, the present note works with fields. If one later wishes to allow structures in which nonzero elements need not be invertible, the natural replacement is an integral domain or commutative ring with identity. The current properness criterion would then need modification, or it would need to be applied after passing to a fraction field.

8. Conclusion

The framework can be stated cleanly as follows: first construct a field extension E/F together with its extra operation $\oplus$; next extend a differential function field (K, D) to $(J, \tilde{D})$ by preserving a declared package of overarching axioms; finally test the explicit coverage and branch conditions for $\Phi(r, k) = r\tilde{g}(\iota k)$. The differential condition $D(g) = g$ is preserved by a differential extension, whereas the proper-two-form property is controlled by the separate multiplicative coverage condition $E^* = F^* \tilde{g}(\iota A)$.

References and source framework

- E. R. Kolchin, Differential Algebra and Algebraic Groups, Academic Press, 1973.

- M. van der Put and M. F. Singer, Galois Theory of Linear Differential Equations, Springer, 2003.
- S. Lang, Algebra, revised third edition, Springer, 2002.
- Framework of n-Overarching Axioms: Overcomplete Forms and Generalized Parameterizations, supplied manuscript, 2026.
- Generalized Complex Field Exploration, supplied exploratory notes, 2026.